

How do sovereign wealth funds pay their firm executives?

Abstract

Sovereign wealth funds (SWFs) are major players in the global markets. We contribute to the corporate governance literature by examining the possible value SWFs bring to their holdings. We specifically examine the impact of SWFs' ownership on firm executive compensation. Using data on Kuwaiti SWFs, we find that having a SWF as an ultimate owner enhances the pay-performance sensitivity to levels matching those in more developed markets. This pay-performance enhancement is greater, the greater the cash flow rights of the SWF is. However, the further away the firm is from the SWF along the ownership chain, as captured by the divergence in cash flow and control rights, diminishes significantly SWFs positive impact. This evidence supports the notion that SWFs create value to their targets through activism, monitoring, and corporate governance enhancements.

JEL classification: *G15, G23, G32, G34, G38*

Keywords: *Executive compensation; Firm performance; Ownership structure; Sovereign wealth funds; Corporate governance*

I Introduction

Sovereign wealth funds (SWFs) are major players in today's global markets holding shares in one out of every five firms worldwide and accounting for about 2% of the global equity and bonds markets ([Fernandes, 2011](#); [Gieve et al., 2009](#)). They are quite distinct from other institutional investors due to their state ownership, mixed objectives, and, for many, no liability structure ([Megginson and Fotak, 2015](#)). Lately, there has been significant debate and disparate findings as to whether SWFs are active investors that create or destroy value for their investments or if they are simply passive investment managers ([Alhashel, 2015](#)). In this paper we intend to examine one possible effect for SWFs on their firm holdings, that is the firm's corporate governance. Specifically, we intend to examine how SWFs might enhance or diminish the pay-performance relationship for the firm's executives.

Executive compensation is a very important element of corporate governance that incentivizes managers to work in the interest of shareholders and reduce agency costs ([Berle and Means, 1991](#); [Jensen and Meckling, 1976](#); [Jensen and Murphy, 1990](#)). Given their position as one of the largest shareholders of the firm, if not the largest, SWFs have a huge monitoring incentive and influence on managerial pay ([Core and Guay, 1999](#); [Jensen and Warner, 1988](#); [Murphy, 1999](#)). This is achieved through their tendency to sit on their firms' boards and/or the corporate governance enhancement proposals they put forward ([Alhashel, 2015](#)). In fact, large and institutional shareholders have been documented to

positively affect the pay performance relationship (Hartzell and Starks, 2003; Ozkan, 2007; Sapp, 2008). However, recent evidence indicates that the influence of the largest shareholder on executive pay differs based on this shareholder's type (Firth et al., 2006). Furthermore, depending on the ownership structure, large shareholders might not always set proper incentive schemes for managers (Wang and Xiao, 2011).

This capacity to influence firm decisions is much greater in economies with concentrated ownership where it is the largest shareholder, or shareholders, that dominate the board, control the firm and, consequently, set executive pay (Firth et al., 2006). Yet, in such economies and with concentrated ownership along comes the divergence of the cash flow and control rights of the controlling shareholders that is harmful to firm value (Claessens et al., 2002). This divergence allows large shareholders to influence firm decisions through voting rights despite relatively smaller cash flow rights. Moreover, it allows majority shareholders and managers to expropriate from the firm at the cost of minority shareholders (Chen et al., 2011). One possible way for managerial rent seeking in such a setting is to be able to receive pay that is not commensurate with their performance (Cao et al., 2011). This is more true for economies with weak investor protection (La Porta et al., 1999; Johnson et al., 2000). Therefore, given the distinct features of SWFs, their large ownership stakes in firms, and influence large shareholders have, and the weakly governed economies they operate in, it becomes important to understand the impact they have on executive pay.

Using data on Kuwaiti firms over the period 2004-2012, we find that in

firms where the ultimate shareholder is a SWF, managers have a higher pay performance sensitivity. SWFs' ownership increases the pay performance sensitivity by 4-5 folds from its base levels, a very substantial increase. Such a considerable effect brings great value to the firm by reducing its agency costs. This higher sensitivity is only true when using accounting-based measures and not with a market-based performance measure. This positive influence that SWFs bring to managerial pay performance sensitivity is greater, the greater the cash flow rights of the SWF are. SWF's positive influence is again found considerable as a one standard deviation increase in their cash flow rights increases the pay performance sensitivity by three folds. However, the positive value SWFs bring in aligning managerial pay and shareholder interests diminishes as the control and cash flow rights of the SWF diverge.

One possible concern with these findings is that SWFs do not bring monitoring value to their holdings but are instead attracted to properly run firms with good corporate governance systems (i.e. clientele effect) ([Brickley et al., 1988](#); [Allen et al., 2000](#); [Bushee and Noe, 2000](#)). As will be discussed in more detail later, this is not the case with our chosen SWFs, except for a handful of instances, as the ownership of the SWF was there since the establishment of the firm and not later. Therefore, we would argue that it is the monitoring of the SWF that brought good governance to the firm and not the other way around.

This paper contributes to several strands of the literature. First of all, it contributes to the institutional investors literature and notably to the growing literature on SWFs. It specifically contributes to the ongoing

debate on whether SWFs create or destroy value in their target investments ([Alhashel, 2015](#); [Megginson and Fotak, 2015](#)). By finding that SWFs improve the pay performance relationship, we provide evidence supporting the notion that SWFs bring value to their targets through their activism and corporate governance enhancements.

Second, it contributes to the literature on executive compensation and ownership structure. While several papers have examined the effect of blockholder, blockholder type, and various ownership structures on executive pay and the pay performance relationship (e.g. [Masulis et al., 2009](#); [Kato and Long, 2006](#); [Firth et al., 2006](#)), this is the first paper to investigate the effects of the unique type of SWFs on executive pay. The paper also contributes to the literature by showing how depending on the owner's incentives and market structure, non-market based performance measures could be used as benchmarks for managerial performance.

Last but not least, this paper contributes to the growing literature on executive compensation in emerging markets. Given the unique setting of developing nations that is different from that observed in the US or Western Europe, several calls have been made into single country research on corporate governance to enhance our understanding of it ([Defond and Hung, 2004](#); [Kanagaretnam et al., 2009](#); [Sun et al., 2010](#)). Kuwait with its large and old SWFs, weak investor protection, ownership concentration, and nascent corporate governance rules, makes it a good laboratory to test for the impact of distinctive entities such as SWFs on firms.

II Institutional background

Kuwait is home to the first and oldest SWF, Kuwait Investment Authority (KIA), which dates its origins back to 1953¹. KIA was established to manage the government surpluses generated from the oil revenues. It invests those funds locally and internationally. The Sovereign Wealth Funds Institute estimates that KIA has assets under management around US\$592 billion as of end of 2016².

Kuwait has also other funds that can also be considered as SWFs, albeit at a much smaller size. The second largest fund is the Public Institute for Social Security (PIFSS) which manages the public pension fund. The third fund is the Public Authority for Minors' Affairs (PAMA) which manages the assets for Kuwaiti minors until they reach legal age. The fourth fund is the Kuwait Awqaf Public Foundation (KAPF). KAPF manages assets for the purpose of spending the returns and proceeds generated from these assets on charitable projects and endeavors. The assets of KAPF are accumulated through new donations or the organic growth of previous donation. Hence, KAPF is similar in structure to non-profit foundations found in the US and else where. The difference, however, is that KAPF is a government entity and not a private one. All of those funds have assets that vary, ranging from equities to alternative assets and from domestic assets to international ones.

These SWFs equity investments in the domestic market tend to be done through two channels. The first channel is through investing some of their funds with professional asset managers who have great if not full discretion

¹ABOUT KIA Overview. <http://www.kia.gov.kw/en/ABOUTKIA/Pages/Overview.aspx>

²<http://www.swfinstitute.org/swfs/kuwait-investment-authority/>

on the allocation and investment of these funds. The second channel is direct equity ownership in local firms that tends to be of significant size (i.e. $\geq 5\%$). In the second channel this direct investment is almost always done from the establishment of the firm. In very few instances does one of those funds, over our sample period, acquire a stake in the firm at a later stage of its life. For example, Kuwait Finance House (KFH), one of the largest and oldest Islamic banks in the world, has been owned jointly by KIA, PAMA, and KAPF with their ownership totaling about 40% since the bank's establishment back in 1977. One reason behind KIA's such channel of investing is Kuwaiti laws that require that when establishing any state-owned enterprise that KIA and other SWFs own at least 25%. The remaining 75% are divided between the general public, 50%, and a major private firm, 25%. This is true for firms in which SWFs have direct ownership and does not apply to firms down the ownership chain.

In those firms where the SWFs own a majority (i.e. $\geq 5\%$), they would tend to have seats on the board depending on their ownership stake. Those seats would usually tend to be filled by those funds' higher management. Through those seats and voting rights, the SWFs, as a dominant shareholder, have a significant say in choosing the top management team, setting their pay, and managing the firm. However, it could be the case that the SWF is not the only large shareholder of the firm and there might be other large shareholders that would also sit on the firm's board of directors.

KIA has been going for the last thirty years through a slow program of divesting its ownership in some of its public ownerships. This is usually done

by offering those shares for the public to bid for at a predetermined price. In very few cases there would be an auction and KIA or the other funds would sell their block ownership to another large investor in the auction.

III Hypothesis Development

SWFs are institutional investors that are quite unique and different from other kinds of institutional investors ([Megginson and Fotak, 2015](#)). They are state-owned investment vehicles funded by foreign exchange reserves, commodity revenues, government budget surplus, or pension surplus that invest in various asset classes globally ([Alhashel, 2015](#)). Furthermore, they tend to have mixed objectives and are not limited to the common financial objective of maximizing the managed wealth. SWFs could be called upon to support their domestic economies, acting in some cases as investors and lenders of last resort, as well as possibly having the mandate to generate social returns, not just economic returns ([Bernstein et al., 2013](#); [Dyck and Morse, 2011](#)). An additional feature of SWFs is their ability to be very long-term investors. For example, KIA has owned shares in Daimler, the owners of Mercedes-Benz, since 1974. This is not constrained to international investments only. KIA has owned shares in Zain, the largest telecommunications operator in Kuwait, since its establishment in 1983.

SWFs, as institutional investors, could be able to bring value to their target investments through their monitoring, activism, and board representation ([Shleifer and Vishny, 1986](#); [Smith, 1996](#)). This is further augmented by their very long investment horizons and very low liquidity

needs which could allow them to reach high and effective monitoring levels (Chen et al., 2007). This high monitoring level would result in a better alignment of managerial and shareholder interests prone to blockholders as captured by evidence of higher pay-performance sensitivity (Ke et al., 1999; Kim, 2010; Hartzell and Starks, 2003; John et al., 2010). This high monitoring is not without its costs and yet it is large long-term investors such as SWFs that are able to afford it (Demsetz and Lehn, 1985; Khan et al., 2005; Shleifer and Vishny, 1986).

However, being state-owned entities, could eventually limit their monitoring of firm managers. This could be driven either by the rent-seeking of politicians (Megginson and Fotak, 2015; Shleifer and Vishny, 1994; ?), a detachment from the firm that is prone in states (Shleifer, 1998), or the pursuit of social goals. If the firm was required to achieve non-financial social objectives (e.g. employment), then managers would not be paid according to their financial performance but based on those non-financial social targets.

The empirical evidence on SWFs' role seems to be mixed. On one hand, there are some papers that find a premium for SWFs investments with the premium stemming mostly from better governance and activism (Dewenter et al., 2010; Fernandes, 2014; Sojli and Wah Tham, 2011). On the other, there have been others showing no value added or, in some cases, value destruction (Kotter and Lel, 2011; Knill et al., 2012; Bortolotti et al., 2015). Therefore, we hypothesize the following,

H 1 *SWFs have no effect on the pay performance sensitivity of manager*

pay.

In environments with weak investor protection, lack of proper corporate governance rules, and abundant use of cross-holdings and pyramid structures, large shareholders have greater incentives to monitor managers and insure their interests are aligned. This monitoring incentive would increase as the amount of cash flow rights the largest or ultimate shareholder has increases (Cao et al., 2011). However, as the cash flow and control rights diverge, the monitoring incentives for large shareholders are weakened and instead those shareholders are found to pursue their own private interests at the expense of other shareholders. Such expropriation would occur through tunneling, related party sales, and transferring profits out of the company for personal gains (Fan et al., 2011; Johnson et al., 2000). This is more true when firms have weak corporate governance mechanisms and disclosure requirements (Ding et al., 2007). While such expropriation incentives might not be there for SWFs, they could certainly be of interest to the other large shareholders who actively manage the firm through their board seats. SWFs, hence, could act as monitors not just of firm managers but of other majority shareholders ensuring they do not expropriate from the firm at the expense of the SWF and the minority shareholders. However, with the firm so far along the ownership chain, it might be hard for the SWF to ensure other majority shareholders are not expropriating the firm, the managers are not colluding with other majority shareholders to extract rents, and/or that managers are paid in a manner commensurate with their performance (i.e. properly monitored) (Cao

et al., 2011; Wang and Xiao, 2011; Cheung et al., 2006; Jiang et al., 2010).

Therefore, we hypothesize the following,

H 2 *SWF's cash flow rights have a positive effect on the pay performance sensitivity of manager pay.*

H 3 *SWF's cash flow control rights divergence has a negative effect on the pay performance sensitivity of manager pay.*

IV Data and Methodology

i Sample

Our data consists of data on publicly listed firms on the KSE over the period 2004-2012. 2004 is the year firms started reporting in their annual reports the top executives' compensation figures. In order to compile our database, we first started by hand collecting executive compensation information directly from the annual reports. Ownership information is provided by Aljoman Consulting. Financial information is obtained from the National Investment Company Investor Pocket Guide and Worldscope. Information on market values is provided through Alshall Economic Consultants. The sample excludes firms in the banking sector given their different characteristics and regulations. Following the combining of those datasets, we arrive at a final sample comprising 169 firms and 906 firm-year observations.

ii Variable definitions

Executive compensation

Kuwaiti firms have been required to report executive compensation in their annual reports' footnotes since 2004. The information provided on compensation is (a) salaries and other short-term benefits, (b) other benefits, and (c) total compensation. The other benefits category is only used by a small number of firms. No information is provided on the nature of the other benefits categories and is only reported for about 10% of firm-year observations in our sample. Given its unclear nature and the scarcity of its use, we focus instead on the cash and short-term benefits category. Such a focus would go along with the extant literature on emerging markets examining mostly cash pay (Chen et al., 2010; Firth et al., 2006, 2007; Cao et al., 2011; Conyon and He, 2011). Moreover, this focus on cash and short-term benefits goes along with the norm in KSE firms in which stock options are not a dominant form of compensation. The few firms that do use stock options do not disclose it clearly with some firms using the term *Other compensation* which may include other items unclear to us³.

The information on executive compensation is provided for the whole top management team and is not reported separately by executive. We take the top management aggregate value of the salaries and other short-term benefits category as our measure of executive compensation. Since no information is provided in the annual reports on the number of executives

³We have done our analysis taking into account both short-term compensation and others and have arrived at the same results

included in the compensation figures, we refrain from examining levels of pay and focus instead on pay-performance relationships only.

Firm performance

In order to estimate the pay-performance sensitivity (PPS), we need to tie changes in pay to changes in firm performance. PPS, hence, would be the the dinar change in executive compensation given a dinar change in firm performance⁴ We use three different firm performance measures based on accounting and market performance. The use of three different measures is driven by the possible inclination some boards might have towards market performance against accounting performance, or vice versa. We first start by examining a market-based measure as captured by the change in total firm market value ([Jensen and Murphy, 1990](#); [Murphy, 1985](#)). Such a measure would tie directly to the increase in shareholders wealth, is less open to managerial manipulation, and reflect a longer-term view of the firm.

A downside, however, for the use of changes in market cap is the possibility that the informational efficiency of stock prices is weak or not strong enough and would vary significantly across different firms ([Abdmoulah, 2010](#); [Al-Ajmi and Kim, 2012](#); [Alhashel, 2016](#)). Moreover, changes in market cap could be affected by factors beyond managerial control such as macroeconomic shocks. The use of accounting-based performance measures is further needed given that bonuses are usually tied to accounting profitability ([Murphy, 1999](#)). The need to use accounting-based measures is also driven by an institutional feature of

⁴Kuwait's currency is the dinar.

SWFs in Kuwait. All SWFs are monitored by the State Audit Bureau (SAB), an independent organization that audits all government entities and ensures the proper use of public funds. In performing this role, SAB annually produces a report regarding all governmental entities in Kuwait that includes the SWFs, their major investments, and the overall performance of the fund as well as the individual investments or firms. The annual report is submitted to and discussed at length in the parliament and any red flags raised by SAB could potentially lead to a drilling of the responsible government minister. In assessing the performance of SWFs' investments, SAB would in many instances focus on the accounting performance of the target firms. Therefore, this would push the SWFs to ensure the proper accounting performance of their holdings. Last but not least, not all SWFs representatives in board rooms come from a finance and investment background, as some could come from legal backgrounds. Given their non-finance backgrounds, some SWF representatives might push for performance measures that they would feel better acquainted with.

To properly address these issues and to capture the possible variation across board rooms in setting performance targets, we augment the market measure with two accounting-based measures, change in sales and change in net income (Core and Guay, 1999). We look at sales given its better reflection of the operating performance of the firm as net income could be more susceptible to accounting manipulation as well as including non-core earnings such as asset sales. We look at contemporaneous changes in our firm performance measures when assessing the PPS for all

of our specifications.

SWFs, cash flow rights and control rights

In assessing the effect of SWFs on executive pay, we determine the ultimate controlling shareholder by tracking it across the ownership chain. We define a major ultimate owner as any shareholder ultimately owning 10% or more of the firm's shares ([Cao et al., 2011](#); [Claessens et al., 2002](#)). In this paper we are interested in understanding the effect of the unique ownership of SWFs. We define SWF ownership as a firm ultimately owned by one of the four SWFs operating in Kuwait; KIA, PIFSS, KAPF, and PAMA. In the instances two or more of those funds are major owners of the firm, we sum their ownership and consider them all as a single entity. This would fit as all four funds are state-owned with some being chaired by the same minister, such as KIA and PIFSS, which would create similarity in their goals, mandates, and behavior. KIA is by far the largest with significant assets under management putting it amongst the largest SWFs in the world ([Alhashel, 2015](#)).

We also examine the effects of the divergence of control and cash flow rights on the pay-performance relationship. We define control rights as the minimum level of ownership in the ownership chain of the ultimate major shareholder. Cash flow rights are defined as the product of the ownership levels along the ownership chain ([La Porta et al., 1999](#); [Claessens et al., 2002](#)). The divergence between the two is calculated as control rights minus cash flow rights. The divergence between cash flow and controls rights is an important feature of ownership structures as it measures the degree of

entrenchment, agency problems, and value destruction in the firm (Claessens et al., 2002; Grossman and Hart, 1986). Such a divergence would emerge when ownership is based on pyramid structures and cross-holdings (La Porta et al., 1999). To get a sense of how cash flow and control rights are measured, let's assume firm A has a 35% ownership in firm B. Firm A in turn is owned by a private family firm where they hold 7% of firm A's shares. The family firm's cash flow rights would be $35\% \times 7\% = 2.45\%$ and their control rights would be the minimal shareholding level across the ownership chain which is 7%.

iii Sample statistics

Table I reports the summary statistics for our sample. We can see that the average top management team (TMT) receives almost KWD 590,000 in cash pay annually which corresponds to about USD 2 millions given the exchange rate observed during the sample period (KWD 1 $\approx$ US\$ 3.6). We can also see from the table that the average firm has a Tobin's Q of 1.106 and a leverage of about 28%. The ultimate owner in the average (median) firm has cash flow rights of about 21% (14%) and a cash flow control rights divergence of about 4% (0%). This would indicate that more than half of the ultimate owners have direct ownership in the firm and not through cross-holdings and pyramid structures. SWFs are the ultimate owners in about 14% of the firm-year observations.

Table II shows us the evolution of pay over the sample years. We can see that pay continued increasing until 2007 after which it dropped following the significant drop in the market at end of 2008 which continued well into

2009. Following 2008 pay almost remained flat in terms of averages except for 2010. [Table III](#) shows the distribution of pay across the KSE sectors. We can clearly see that there are big variations across the industries with the Insurance (Investment) sector having the lowest (highest) pay. The variation in pay across industries, however, is less when looking at the median pay. This is most likely due to the significant firm size variation that is observed within some industries.

[Table IV](#) provides insight into proportion of firms with a SWF as an ultimate owner. We can see that the SWFs' ultimate ownership by 2012 was half the ownership levels they had at the beginning of the sample period. Recall that these are instances in which the SWF is the ultimate owner with no less than 10% control rights in the firm. This could be driven by two elements. First, and possibly the stronger effect, is the significant increase in the number of listed firms over the sample period especially by small firms that are not owned by the SWFs. Second, KIA exiting from several major ownership stakes, either fully or partially.

[Table V](#) shows the correlation matrix of our variables. We can see, as is heavily documented in the extant literature, that pay is highly correlated with firm size, whether measured in terms of market cap, total assets, or sales.

V Empirical results

In this section we examine the effect of having a SWF as the ultimate owner on the pay-performance relationship for the firm's executives. We start by

examining the effect of the mere presence of a SWF and then expand by taking into account its cash flow rights. Our last set of analysis intends to test the hypotheses regarding the cash flow rights and the cash flow control rights divergence of the SWF on the pay-performance sensitivity.

i SWF and pay-performance relationship

Our first step is to examine the pay-performance relationship in Kuwaiti listed firms while taking into account the impact of having a SWF as an ultimate owner. We do that by estimating the following model,

$$\begin{aligned} \Delta Pay_{it} = & \beta_o + \beta_1 \Delta Perf_{it} * SWF_{it} + \beta_2 \Delta Perf_{it} + \beta_3 SWF_{it} + \beta_4 X_{it} \\ & + \gamma_i + \zeta_t + \epsilon_{it} \end{aligned} \tag{1}$$

where i and t represent firm and year, respectively, ΔPay is the change in pay as defined earlier, $Perf$ is one of the performance measures discussed earlier and $\Delta Perf$ is the change in this performance measure, SWF is a dummy variable equal to one if the largest shareholder is a SWF and zero otherwise, X are control variables, γ is the industry fixed effect, ζ is the year fixed effect, and ϵ is the error term. All errors are clustered at the firm level (Petersen, 2009). By examining the change in compensation relative to change in performance, as in Equation (1), we would be able to assess the degree of alignment in the interests of the managers and the shareholders (i.e. the pay performance sensitivity) (Conyon and Murphy, 2000; Coughlan and Schmidt, 1985; Jensen and Murphy, 1990; Murphy, 1985). Furthermore, sensitivity specifications can easily be interpreted economically and would

tell us in a straight forward fashion the managerial share of the value creation (Murphy, 1999). Additionally, change specifications allows us to control for CEO-specific factors and eliminate time-invariant firm fixed effect (Conyon and He, 2011; Jackson et al., 2008; Murphy, 1999).

Similar to Hartzell and Starks (2003), Firth et al. (2006), Conyon and He (2011) and Cao et al. (2011), we control for Tobin’s q , firm size, and leverage. Firm size, measured as the market cap of the firm, has extensively been documented as related to executive pay and the pay-performance relationship and the level of institutional ownership in the firm (Murphy, 1999; Conyon, 1997; Baker and Hall, 2004; Schaefer, 1998; Sias and Starks, 1997). Tobin’s q , a proxy for growth opportunities, has been shown to affect managerial incentives. Smith and Watts (1992) and Harvey and Shrieves (2001) argue that firms with greater growth opportunities are harder to monitor and hence it becomes more essential to use pay incentive mechanisms to align the interests of shareholders and managers. We define Tobin’s q as total market capitalization plus debt divided by total book value of assets. Leverage, defined as debt divided by the book value of assets, can affect the structure of compensation contracts used by the firm to better align the interests of managers and debtholders (John and John, 1993). The use of industry fixed effects is to capture any industry-specific practices in terms of compensation, industry-specific managerial talents, and/or SWF’s preference for certain industries (Lippert et al., 1994). Table III shows a list of the industries in our sample

following the KSE classification ⁵ Year fixed effects are used to capture any macroeconomic shocks to the whole market.

The Equation (1) estimate are shown in Table VI for the three different performance measures. The table indicates that there exists a pay-performance relationship on average as evidenced by the significant and positive coefficients on $\Delta MarketCap$ and $\Delta NetIncome$. The evidence indicates that KWD 1,000 change in net profit increases top managerial pay by KWD 5. We also find that larger firms have greater pay-performance sensitivities as documented in the literature.

More importantly, we can see that ultimate ownership by a SWF has a positive and significant effect on the pay-performance relationship. This is evidenced by the significant coefficients for the interaction terms in columns 2 and 3. However, we fail to find an effect of SWF ownership on the pay-performance sensitivity when performance is measured using changes in market value. Having a SWF be the ultimate owner increases the pay-performance sensitivity from 0.001 (0.005) to 0.005 (0.029) if performance is measured using change in sales (net income). Those are very noteworthy increases in the pay-performance sensitivity relative to the baseline sensitivity as captured by the coefficients on the performance measures. The findings also point out that the SWFs prefer to use accounting-based performance measures to pay executives as opposed to market-performance.

To get a better sense of these pay-performance elasticities, we compare

⁵The sector classification is currently different. However, we have opted to use the original classification that was prevalent during most of the sample period.

our figures with those reported for the US. [Murphy \(1999\)](#) reports elasticities ranging between 0.005-0.014 for the cash pay-shareholder wealth relationship for firms in the early 90s. Our estimates for SWFs, ranging between 0.005-0.029, would put them in line with those found in the US. In other words, having the SWFs as ultimate owners raises the pay-performance sensitivity to that documented in highly developed markets.

The results from [Table VI](#) support the hypothesis that SWFs add value to firms through their active board roles and corporate governance enhancements.

ii SWF cash flow rights, and pay-performance relationship

In this section we care to look at the size of the ultimate owner’s incentives by examining his cash flow rights instead of a dummy variable as in the previous section. This allows us to further understand the effect of SWFs ownership on the pay-performance relationship as their incentives in the firm changes. We adjust our previous specification in a vain similar to that of [Firth et al. \(2006, 2007\)](#), [Canarella and Nourayi \(2008\)](#), and [Cao et al. \(2011\)](#). It is, then, as follows,

$$\begin{aligned} \Delta Pay_{it} = & \beta_o + \beta_1 \Delta Perf_{it} * SWF\ Cash_{it} + \beta_2 \Delta Perf_{it} \\ & + \beta_3 Cash\ flow\ rights_{it} + \beta_4 X_{it} + \gamma_i + \zeta_t + \epsilon_{it}. \end{aligned} \quad (2)$$

where *SWF Cash* is the cash flow rights of the SWF and *Cash flow rights* is the cash flow rights of the ultimate owner.

The estimates are shown in [Table VII](#) broken down by our three

performance measures. As we had established in the previous table, there is a positive and significant relationship between performance and pay as captured by changes in market cap and changes in net income. We now move on to look at the interaction terms between performance and the cash flow rights of SWFs. We can see that as SWFs' cash flow rights increased, the pay-performance relationship is strengthened. However, as in [Table VI](#), this is only true if performance is measured using changes in sales and net income. This augmentation that SWFs bring to the pay-performance elasticity statistically significant at the 1% and 10% for changes in sales and changes in net income, respectively. A one standard deviation increase in SWF's cash flow rights, 0.2, results in an increase in the pay-performance elasticity of $0.001 + 0.2 * 0.015 = 0.004$, a 300% increase in the base elasticity. Similar magnitude increases are observed for similar changes in SWF cash flow rights when performance is measured using change in net income.

Overall, the results support the notion that the greater the cash flow rights of SWFs, the greater the benefits they bring to the firm through paying managers commensurately with their performance.

iii SWF, ownership structure, and pay-performance relationship

In this section we examine the intersection of SWF ownership and the structure they utilize to hold the firm's shares. Essentially, we try to understand how pay-performance sensitivity is affected by pyramid structures or cross-holdings in which the ultimate owner residing at the

top of the chain is a SWF. Here we utilize our earlier specification, Equation (2), but use cash flow control rights divergence in place of cash flow rights as follows,

$$\begin{aligned} \Delta Pay_{it} = & \beta_o + \beta_1 \Delta Perf_{it} * SWF CCD_{it} + \beta_2 \Delta Perf_{it} + \beta_3 CCD_{it} \\ & + \beta_4 X_{it} + \gamma_i + \zeta_t + \epsilon_{it}. \end{aligned} \quad (3)$$

where *SWF CCD* is the cash flow control rights divergence of SWF and *CCD* is the cash flow control rights of the ultimate shareholder. The rest of the variables are as defined previously.

Table VIII reports the estimates of Equation (3). As in our earlier tables, we find that a pay-performance relationship exists that is statistically significant for the average firm across all three performance measures. We can see that in both columns 1 and 2 the greater the divergence in rights of SWFs the worse is the pay-performance relationship as evidenced by the negative coefficients on $\Delta MV * SWF CCD$ and $\Delta Sales * SWF CCD$. The effect is more statistically significant in column 1 at 5% than in column 2. To get a sense of how severe is the negative coefficient, we observe how a one standard deviation change in SWF CCD changes the total pay-performance sensitivity. A one-standard deviation increase in SWF CCD (0.083) would result in the pay-performance sensitivity reaching $0.003 + 0.083 * -0.010 = 0.002$ when performance is measured using change in market cap. When performance is measured as changes in sales the pay-performance sensitivity goes down to $0.004 + 0.083 * -0.132 = -0.008$. The effect, therefore, is so severe that it completely

washes away the positive and significant coefficient on the performance measures. Essentially, even though SWFs had a positive effect on the pay-performance relationship as we saw in [Table VI](#), this positive effect is diluted as the SWF moves upward in the ownership chain until it is completely eliminated or worse reversed. A pay-performance sensitivity of zero or less would mean that executives are not compensated in a manner commensurate with their performance and might be able to extract rewards even when they perform poorly.

To summarize, we find evidence in support of our hypothesis that a greater divergence between the cash flow and control rights of SWFs result in a weaker pay-performance relationship. In other words, the further away the ultimate owner is in the ownership chain or structure, the weaker is the monitoring of management. This finding goes along with the extant literature that the separation of ownership and control fosters weaker corporate governance and, hence, allowing managers and/or majority shareholders to enjoy rents at the expense of shareholders ([Claessens et al., 2002](#); [La Porta et al., 1999](#)).

VI Conclusion

The rise of SWFs over the last decades has attracted significant attention from both academics and practitioners. This was especially true at the dawn of the 2008 financial crisis when several SWFs acted as lenders and investors of last resort for some of the largest global banks. The unique structure and objectives of SWFs, in comparison to other institutional

investors, has been a major impetus in trying to understand them. In this paper we tried to contribute to the literature’s understanding of SWFs as investors and the potential value they could bring to or destroy in their firm holdings. We tackle this by examining the effect SWFs have on a very central element in enhancing firm value and that is executive compensation in their firm holdings. Specifically, we investigate their effect on the alignment of managerial pay and performance in Kuwaiti listed firms.

We show that SWFs bring significant enhancements to the alignment of executive pay to performance. Having a SWF increases the pay-performance sensitivity by four folds, a very notable increase. The findings show that SWFs have an influence on the pay performance relationship when performance is assessed on an accounting basis. This influence is greater the higher the cash flow rights of the SWF in the firm are. However, similar to the evidence in the extant literature on the negative effects of control rights cash flow rights divergence, we find that such a divergence in SWF’s ownership hinders the pay-performance sensitivity. The negative effect is so severe that it could eliminate any pay performance relationship existing in the firm.

We show that, based on the investment behavior of the SWFs in Kuwait, the positive effect we find is not driven by SWF being attracted to well-governed firms (i.e. clientele effect). Instead, the SWFs have been major shareholders of the firm since the firm’s inception and, hence, it is the SWF that brought the value enhancement.

The results indicate that SWFs bring value to their target holdings through their board activism, corporate governance enhancements, and

monitoring even though state-owned entities have been documented to fail in this regard. The findings indicate that SWFs use incentive compensation and directly tie managerial pay to firm performance to reduce agency costs. This conclusion goes to support the previous findings in the literature on the value of SWFs' activism ([Dewenter et al., 2010](#); [Fernandes, 2014](#); [Sojli and Wah Tham, 2011](#)). However, a detachment of the SWF from its target holdings through pyramid structures significantly weakens their monitoring role and allows for the possible rent extraction from the firm by other majority shareholders and/or managers.

The finding that SWFs bring value to their firms through better management monitoring should promote additional research on other value enhancement mechanisms SWFs could put forward. Those could range from the proper management of a firm's cash holding and investments to its dividend policy. Future research could also try to understand whether SWFs behavior and activism depends on whether their target holdings were domestic or international. Such a difference across countries could arise due to the mixed objectives SWFs might have with their domestic investments. Moreover, SWFs behavior in international markets could differ based on the host country's reception of foreign investments and the type of diplomatic relations the SWF's country and the host have.

References

- Abdmoulah, W. (2010). Testing the evolving efficiency of arab stock markets. *International Review of Financial Analysis*, 19(1):25–34.
- Al-Ajmi, J. and Kim, J. (2012). Are gulf stock markets efficient? evidence from new multiple variance ratio tests. *Applied Economics*, 44(14):1737–1747.
- Alhashel, B. (2015). Sovereign wealth funds: a literature review. *Journal of Economics and Business*, 78:1–13.
- Alhashel, B. S. (2016). Rights offering announcements and the efficiency of the kuwaiti market. *Applied Economics Letters*, pages 1–5.
- Allen, F., Bernardo, A. E., and Welch, I. (2000). A theory of dividends based on tax clienteles. *The Journal of Finance*, 55(6):2499–2536.
- Baker, G. P. and Hall, B. J. (2004). Ceo incentives and firm size. *Journal of Labor Economics*, 22(4):767–798.
- Berle, A. A. and Means, G. G. C. (1991). *The modern corporation and private property*. Transaction publishers.
- Bernstein, S., Lerner, J., and Schoar, A. (2013). The investment strategies of sovereign wealth funds. *The Journal of Economic Perspectives*, 27(2):219–238.
- Bortolotti, B., Fotak, V., and Megginson, W. L. (2015). The sovereign wealth fund discount: evidence from public equity investments. *Review of Financial Studies*, page hhv036.
- Brickley, J. A., Lease, R. C., and Smith, C. W. (1988). Ownership structure and voting on antitakeover amendments. *Journal of financial economics*, 20:267–291.
- Bushee, B. J. and Noe, C. F. (2000). Corporate disclosure practices, institutional investors, and stock return volatility. *Journal of accounting research*, pages 171–202.
- Canarella, G. and Nourayi, M. M. (2008). Executive compensation and firm performance: adjustment dynamics, non-linearity and asymmetry. *Managerial and Decision Economics*, 29(4):293–315.
- Cao, J., Pan, X., and Tian, G. (2011). Disproportional ownership structure and pay–performance relationship: evidence from china’s listed firms. *Journal of Corporate Finance*, 17(3):541–554.
- Chen, C. J., Li, Z., Su, X., and Sun, Z. (2011). Rent-seeking incentives, corporate political connections, and the control structure of private firms: Chinese evidence. *Journal of Corporate Finance*, 17(2):229–243.
- Chen, J. J., Liu, X., and Li, W. (2010). The effect of insider control and global benchmarks on chinese executive compensation. *Corporate Governance: An International Review*, 18(2):107–123.

- Chen, X., Harford, J., and Li, K. (2007). Monitoring: Which institutions matter? *Journal of Financial Economics*, 86(2):279–305.
- Cheung, Y.-L., Rau, P. R., and Stouraitis, A. (2006). Tunneling, propping, and expropriation: evidence from connected party transactions in hong kong. *Journal of Financial Economics*, 82(2):343–386.
- Claessens, S., Djankov, S., Fan, J., and Lang, L. (2002). Disentangling the incentive and entrenchment effects of large shareholdings. *The Journal of Finance*, 57(6):2741–2771.
- Conyon, M. J. (1997). Corporate governance and executive compensation. *International journal of industrial organization*, 15(4):493–509.
- Conyon, M. J. and He, L. (2011). Executive compensation and corporate governance in china. *Journal of Corporate Finance*, 17(4):1158–1175.
- Conyon, M. J. and Murphy, K. J. (2000). The prince and the pauper? ceo pay in the united states and united kingdom. *The Economic Journal*, 110(467):640–671.
- Core, J. and Guay, W. (1999). The use of equity grants to manage optimal equity incentive levels. *Journal of Accounting and Economics*, 28(2):151–184.
- Coughlan, A. T. and Schmidt, R. M. (1985). Executive compensation, management turnover, and firm performance: An empirical investigation. *Journal of Accounting and Economics*, 7(1-3):43–66.
- Defond, M. L. and Hung, M. (2004). Investor protection and corporate governance: Evidence from worldwide ceo turnover. *Journal of Accounting Research*, 42(2):269–312.
- Demsetz, H. and Lehn, K. (1985). The structure of corporate ownership: Causes and consequences. *Journal of political economy*, 93(6):1155–1177.
- Dewenter, K. L., Han, X., and Malatesta, P. H. (2010). Firm values and sovereign wealth fund investments. *Journal of Financial Economics*, 98(2):256–278.
- Ding, Y., Zhang, H., and Zhang, J. (2007). Private vs state ownership and earnings management: evidence from chinese listed companies. *Corporate Governance: An International Review*, 15(2):223–238.
- Dyck, I. J. A. and Morse, A. (2011). Sovereign Wealth Fund Portfolios. Technical Report 11-15, Chicago Booth Research Paper.
- Fan, J. P., Wei, K. J., and Xu, X. (2011). Corporate finance and governance in emerging markets: A selective review and an agenda for future research. *Journal of Corporate Finance*, 17(2).
- Fernandes, N. (2011). Sovereign wealth funds: Investment choices and implications around the world.
- Fernandes, N. (2014). The impact of sovereign wealth funds on corporate value and performance. *Journal of Applied Corporate Finance*, 26(1):76–84.

- Firth, M., Fung, P. M., and Rui, O. M. (2006). Corporate performance and ceo compensation in china. *Journal of Corporate Finance*, 12(4):693–714.
- Firth, M., Fung, P. M., and Rui, O. M. (2007). How ownership and corporate governance influence chief executive pay in china’s listed firms. *Journal of Business Research*, 60(7):776–785.
- Gieve, J. et al. (2009). Sovereign wealth funds and global imbalances. *Revue d’Économie Financière*, 9(1):163–177.
- Grossman, S. J. and Hart, O. D. (1986). The costs and benefits of ownership: A theory of vertical and lateral integration. *The Journal of Political Economy*, pages 691–719.
- Hartzell, J. C. and Starks, L. T. (2003). Institutional investors and executive compensation. *The journal of finance*, 58(6):2351–2374.
- Harvey, K. D. and Shrieves, R. E. (2001). Executive compensation structure and corporate governance choices. *Journal of Financial Research*, 24(4):495–512.
- Jackson, S. B., Lopez, T. J., and Reitenga, A. L. (2008). Accounting fundamentals and ceo bonus compensation. *Journal of Accounting and Public Policy*, 27(5):374–393.
- Jensen, M. and Meckling, W. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of financial economics*, 3(4):305–360.
- Jensen, M. and Murphy, K. (1990). Performance pay and top-management incentives. *Journal of Political Economy*, 98(2):225.
- Jensen, M. C. and Warner, J. B. (1988). The distribution of power among corporate managers, shareholders, and directors. *Journal of Financial Economics*, 20:3–24.
- Jiang, G., Lee, C. M., and Yue, H. (2010). Tunneling through intercorporate loans: The china experience. *Journal of Financial Economics*, 98(1):1–20.
- John, K., Mehran, H., and Qian, Y. (2010). Outside monitoring and ceo compensation in the banking industry. *Journal of Corporate Finance*, 16(4):383–399.
- John, T. A. and John, K. (1993). Top-management compensation and capital structure. *The Journal of Finance*, 48(3):949–974.
- Johnson, S., La Porta, R., Lopez-de Silanes, F., and Shleifer, A. (2000). Tunneling. *American Economic Review*, 90(2):22–27.
- Kanagaretnam, K., Mathieu, R., and Shehata, M. (2009). Usefulness of comprehensive income reporting in canada. *Journal of Accounting and Public Policy*, 28(4):349–365.

- Kato, T. and Long, C. (2006). Executive compensation, firm performance, and corporate governance in china: Evidence from firms listed in the shanghai and shenzhen stock exchanges. *Economic development and Cultural change*, 54(4):945–983.
- Ke, B., Petroni, K., and Safieddine, A. (1999). Ownership concentration and sensitivity of executive pay to accounting performance measures: Evidence from publicly and privately-held insurance companies. *Journal of Accounting and Economics*, 28(2):185–209.
- Khan, R., Dharwadkar, R., and Brandes, P. (2005). Institutional ownership and ceo compensation: a longitudinal examination. *Journal of Business Research*, 58(8):1078–1088.
- Kim, K. (2010). Blockholder monitoring and the efficiency of pay-performance benchmarking. *Journal of Corporate Finance*, 16(5):748–766.
- Knill, A. M., Lee, B. S., and Mauck, N. (2012). Sovereign wealth fund investment and the return-to-risk performance of target firms. *Journal of Financial Intermediation*, 21(2):315 – 340.
- Kotter, J. and Lel, U. (2011). Friends or foes? Target selection decisions of sovereign wealth funds and their consequences. *Journal of Financial Economics*, 101(2):360–381.
- La Porta, R., Lopez-de Silanes, F., and Shleifer, A. (1999). Corporate ownership around the world. *The journal of finance*, 54(2):471–517.
- Lippert, R. L., Moore, W. T., et al. (1994). Compensation contracts of chief executive officers: Determinants of pay-performance sensitivity. *Journal of Financial Research*, 17(3):321–32.
- Masulis, R. W., Wang, C., and Xie, F. (2009). Agency problems at dual-class companies. *The Journal of Finance*, 64(4):1697–1727.
- Meggison, W. L. and Fotak, V. (2015). Rise of the fiduciary state: A survey of sovereign wealth fund research. *Journal of Economic Surveys*, 29(4):733–778.
- Murphy, K. J. (1985). Corporate performance and managerial remuneration: An empirical analysis. *Journal of accounting and economics*, 7(1):11–42.
- Murphy, K. J. (1999). Executive compensation. *Handbook of labor economics*, 3:2485–2563.
- Ozkan, N. (2007). Do corporate governance mechanisms influence ceo compensation? an empirical investigation of uk companies. *Journal of Multinational Financial Management*, 17(5):349–364.
- Petersen, M. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *Review of Financial Studies*, 22(1):435.
- Sapp, S. G. (2008). The impact of corporate governance on executive compensation. *European Financial Management*, 14(4):710–746.

- Schaefer, S. (1998). The dependence of payperformance sensitivity on the size of the firm. *Review of Economics and Statistics*, 80(3):436–443.
- Shleifer, A. (1998). State versus private ownership. *Journal of Economic Perspectives*, 12(4):133–150.
- Shleifer, A. and Vishny, R. W. (1986). Large shareholders and corporate control. *Journal of political economy*, 94(3, Part 1):461–488.
- Shleifer, A. and Vishny, R. W. (1994). Politicians and firms. *The Quarterly Journal of Economics*, 109(4):995–1025.
- Sias, R. W. and Starks, L. T. (1997). Institutions and individuals at the turn-of-the-year. *The Journal of Finance*, 52(4):1543–1562.
- Smith, C. W. and Watts, R. L. (1992). The investment opportunity set and corporate financing, dividend, and compensation policies. *Journal of financial Economics*, 32(3):263–292.
- Smith, M. P. (1996). Shareholder Activism by Institutional Investors: Evidence from CalPERS. *The Journal of Finance*, 51(1):227–252.
- Sojli, E. and Wah Tham, W. (2011). The impact of foreign government investments: sovereign wealth fund investments in the united states. In *Institutional Investors in Global Capital Markets*, pages 207–243. Emerald Group Publishing Limited.
- Sun, S. L., Zhao, X., and Yang, H. (2010). Executive compensation in asia: A critical review and outlook. *Asia Pacific Journal of Management*, 27(4):775–802.
- Wang, K. and Xiao, X. (2011). Controlling shareholders tunneling and executive compensation: Evidence from china. *Journal of Accounting and Public Policy*, 30(1):89–100.

Table I: Summary Statistics

This table provides summary statistics for the variables used. Variables are as defined in the Appendix. The sample consists of Kuwait non-bank firms over the sample period 2004-2012. All figures are in Kuwaiti Dinar.

	N	Mean	Median	Min	Max	SD
Cash Pay (thousands)	906	593.6	274.2	4	33184	1515.3
Market Cap (millions)	906	144.6	48.82	2.195	7241.4	465.8
Tobin's Q	906	1.098	0.933	0.129	7.310	0.648
Total Assets (millions)	906	191.5	94.60	1.975	5568.2	424.1
Sales (millions)	906	57.09	13.30	-72.29	2003.1	205.7
Net Income (millions)	906	5.969	1.756	-282.0	521.7	40.71
Leverage	906	0.278	0.264	0	1.491	0.204
Cash Flow Rights	906	0.208	0.136	0	0.958	0.202
CCD	906	0.0410	0	0	0.274	0.0675
SWF	906	0.137	0	0	1	0.344

Table II: Executive Pay by Year

This table show the total executive pay annually over the sample period. Variables are as defined in the Appendix. The sample consists of Kuwait non-bank firms over the sample period 2004-2012. All figures are in Kuwaiti Dinar.

	N	Mean	Median	Min	Max	SD
2004	28	433.8	236.6	30	3005.0	612.3
2005	73	595.5	222	19	6338	1022.2
2006	125	464.8	236.3	4	4127	701.7
2007	129	656.2	278	24	7511	1136.1
2008	144	530.7	272.1	5.134	7218	966.5
2009	144	529.5	258.6	21.53	8540	1061.7
2010	149	698.6	231.3	18.24	33184	2809.3
2011	145	543.5	276.3	11.27	9227	1137.8
2012	144	537.6	275.6	17.44	9623	1154.2
Total	1081	565.5	256.8	4	33184	1413.4

Table III: Executive Pay by Industry

This table show the total executive pay by industry. Industry classifications are per the Kuwait Stock Exchange's during the sample period. Variables are as defined in the Appendix. The sample consists of Kuwait non-bank firms over the sample period 2004-2012. All figures are in Kuwaiti Dinar.

	N	Mean	Median	Min	Max	SD
Food	81	504.5	282.9	11.27	4687	767.4
Industrials	175	420.5	201.4	5.134	6798	864.9
Insurance	48	192.7	163.6	78	464.9	91.86
Investment	287	745.9	336.2	4	9623	1308.8
Real Estate	182	369.6	307.2	18.24	2022.2	299.1
Services	308	669.8	217.3	18	33184	2168.3
Total	1081	565.5	256.8	4	33184	1413.4

Table IV: SWF Ownership over Time

This table shows Sovereign Wealth Funds average ultimate ownership size annually. The sample consists of Kuwait non-bank firms over the sample period 2004-2012.

Ownership	
2004	0.261
2005	0.304
2006	0.227
2007	0.163
2008	0.121
2009	0.121
2010	0.122
2011	0.125
2012	0.0979
Total	0.147

Table V: Correlation Matrix

This table shows the correlation between the variables of interest. Variables are as defined in the Appendix. The sample consists of Kuwait non-bank firms over the sample period 2004-2012.

	Cash Pay (thousands)	Market Cap (millions)	Tobin's Q	Total Assets (millions)	Sales (millions)	Net Income (millions)	Leverage	Cash Flow Rights	CCD	SWF
Cash Pay (thousands)	1.000									
Market Cap (millions)	0.662	1.000								
Tobin's Q	0.241	0.214	1.000							
Total Assets (millions)	0.587	0.865	0.009	1.000						
Sales (millions)	0.536	0.602	0.071	0.653	1.000					
Net Income (millions)	0.503	0.676	0.280	0.514	0.577	1.000				
Leverage	0.095	0.034	-0.093	0.170	0.031	-0.089	1.000			
Cash Flow Rights	0.025	-0.007	-0.018	-0.041	-0.015	0.008	0.026	1.000		
CCD	-0.049	-0.085	-0.136	-0.053	-0.001	-0.045	-0.072	-0.380	1.000	
SWF	0.100	0.213	0.078	0.188	0.086	0.151	-0.019	0.020	0.228	1.000

Table VI: Pay-Performance Relationship and Ultimate Owner

This table shows the regression results of change in pay on change on performance and sovereign wealth fund ownership. The dependent variable is change in cash pay. The sample consists of Kuwait non-bank firms over the sample period 2004-2012. Variables are as defined in the Appendix. *t*-statistics are in parentheses. Standard errors are clustered at the firm level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)
Δ Market Cap*SWF	0.000 (1.27)		
Δ Sales*SWF		0.004*** (3.71)	
Δ NI*SWF			0.024*** (5.90)
Δ Market Cap	0.000* (1.96)		
Δ Sales		0.001 (1.49)	
Δ Net Income			0.005*** (3.89)
SWF	-31315.791 (-0.78)	-81643.838 (-0.79)	-31073.720 (-0.33)
Market Cap (millions)	230.021*** (11.06)	383.944*** (5.73)	409.117*** (12.39)
Tobin's Q	43592.058 (0.82)	60597.262 (1.59)	44881.083 (1.06)
Leverage	-222350.563 (-1.11)	-172393.060 (-1.41)	-138175.517 (-1.42)
Obs.	788	804	806
Adj R-Sq	0.259	0.151	0.205
Year FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes

Table VII: Pay-Performance Relationship, Cash Flow Rights, and Ultimate Owner

This table shows the regression results of change in pay on change on performance and sovereign wealth fund cash flow rights. The dependent variable is change in cash pay. The sample consists of Kuwait non-bank firms over the sample period 2004-2012. Variables are as defined in the Appendix. *t*-statistics are in parentheses. Standard errors are clustered at the firm level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)
Δ Market Cap*SWF CF Rights	0.000 (1.28)		
Δ Sales*SWF CF Rights		0.015*** (3.77)	
Δ NI*SWF CF Rights			0.072* (1.95)
Δ Market Cap	0.000** (2.21)		
Δ Sales		0.001 (1.48)	
Δ Net Income			0.006*** (3.85)
Cash Flow Rights	-21758.100 (-0.34)	-8178.965 (-0.13)	-5892.649 (-0.09)
Market Cap (millions)	226.471*** (15.32)	370.522*** (7.48)	440.235*** (10.52)
Tobin's Q	44190.527 (0.84)	63027.815* (1.66)	45346.360 (1.12)
Leverage	-219509.974 (-1.09)	-164084.242 (-1.42)	-157425.916 (-1.57)
Obs.	788	804	806
Adj R-Sq	0.258	0.151	0.179
Year FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes

Table VIII: Pay-Performance Relationship, Ownership Structure, and Ultimate Owner

This table shows the regression results of change in pay on change on performance and sovereign wealth fund control rights cash flow rights divergence. The dependent variable is change in cash pay. The sample consists of Kuwait non-bank firms over the sample period 2004-2012. Variables are as defined in the Appendix. *t*-statistics are in parentheses. Standard errors are clustered at the firm level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)
Δ Market Cap*SWF CCD	-0.000** (-2.50)		
Δ Sales*SWF CCD		-0.132* (-1.74)	
Δ NI*SWF CCD			0.000 (0.01)
Δ Market Cap	0.000*** (14.31)		
Δ Sales		0.004*** (4.54)	
Δ Net Income			0.009** (2.24)
CCD	-201811.336* (-1.97)	52036.264 (0.25)	1464.167 (0.01)
Market Cap (millions)	242.094*** (8.60)	394.050*** (8.36)	536.204*** (14.89)
Tobin's Q	19148.378 (0.35)	49472.309* (1.68)	32788.822 (0.86)
Leverage	-206688.483 (-1.03)	-153096.621 (-1.25)	-178574.611 (-1.56)
Obs.	788	804	806
Adj R-Sq	0.254	0.137	0.120
Year FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes

Appendix: Variable and Acronym Definitions

Variable	Variable Description
<i>Cash Pay</i>	The salaries and other short-term compensation received by the top management team.
<i>Market Cap</i>	The market capitalization of the firm calculated as the stock price multiplied by the number of shares outstanding.
<i>Tobin's Q</i>	Calculated as the market cap of the firm plus its debts divided by the book value of assets.
<i>Total Assets</i>	The total assets of the firm.
<i>Sales</i>	Total sales of the firm.
<i>Net Income</i>	The net income of the firm
<i>Leverage</i>	Calculated as the firm's debt divided by total assets.
<i>Cash Flow Rights</i>	The product of the ownership levels along the ownership chain of the ultimate major shareholder.
<i>Control Rights</i>	The minimum level of ownership in the ownership chain of the ultimate major shareholder.
<i>CCD</i>	The difference between the ultimate shareholder's control and cash flow rights.
<i>SWF</i>	A dummy variable equal to one if the largest ultimate shareholder is a SWF, zero otherwise.